LAB 6: CODE AND BRANCH COVERAGE ANALYSIS

Marathon Registration Application

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1. Objective

The objective of this lab is to test the marathon registration application and measure statement/code coverage and branch coverage using Python, pytest, and coverage.py. The target for both measures is at least 90%.

2. Tools and Environment

- Python 3.13.7
- pytest (test execution)
- coverage.py (code and branch coverage measurement)
- Git and GitHub (version control and submission)

3. Application Under Test

The tests target marathon_registration.py and its three functions:

- entry_fee(race, age, is_early_bird, needs_tshirt) — validates race and age, then calculates the fee.
- start_wave(best_time_minutes, race) — assigns a starting wave based on the runner's best time.
- finisher_status(elapsed_minutes, race, checkpoints_missed) — determines disqualification, finisher, or DNF status.

4. Existing Tests — Before Adding New Coverage Tests

The original test file was located in the lab6 directory. The initial test run collected two tests and both passed. The first attempt from the repository root did not find the file because it was located in the lab6 subdirectory.

Test file	Tests collected	Result
test_marathon_registration.py	2	2 passed

Before:

```
PS C:\Users\Bindushree\Downloads\SE_Lab6\L6_L7_60_marathon_registration\lab6>python -m pytest -v test_marathon_registration.py
========================== test session starts ==========================
platform win32 -- Python 3.13.7, pytest-9.1.1, pluggy-1.7.0 -- C:\Users\Bindushree\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Bindushree\Downloads\SE_Lab6\L6_L7_60_marathon_registration\lab6
plugins: anyio-4.15.1, hydra-core-1.3.7
collected 2 items

test_marathon_registration.py::test_10k_standard PASSED              [ 50%]
test_marathon_registration.py::test_first_timer_wave PASSED          [100%]
```

5. Code Coverage Testing

A separate test file, CodeCoverageTest_AM901_marathon_registration.py, was created to exercise the application functions across valid inputs, invalid race values, age limits, discounts, T-shirt charges, start-wave boundaries, and finisher outcomes.

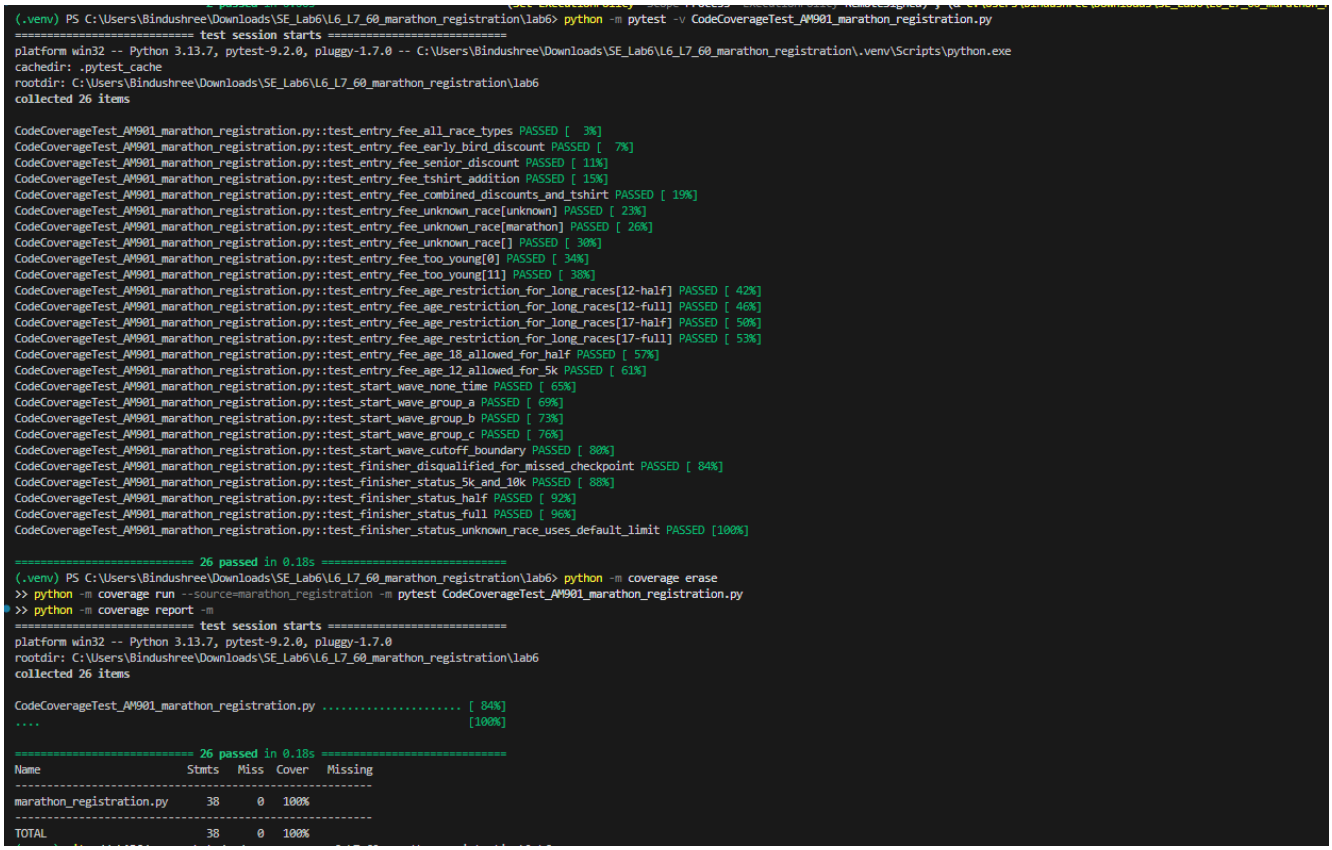
Command Used

python -m coverage erase
python -m coverage run --source=marathon_registration -m pytest CodeCoverageTest_AM901_marathon_registration.py
python -m coverage report -m

Test file	Tests passed	Statements missed	Code coverage
CodeCoverageTest_AM901_marathon_registration.py	26 / 26	0	100%

Result: All 26 code-coverage tests passed. The coverage report showed 38 statements, 0 missed statements, and 100% code coverage.

Code Coverage Result



6. Branch Coverage Testing

A separate test file, BranchCoverageTest_AM901_marathon_registration.py, was created to exercise both outcomes of the application decision conditions, including race validation, age checks, early-bird and senior discounts, T-shirt charges, starting-wave boundaries, checkpoint disqualification, and finish-time limits.

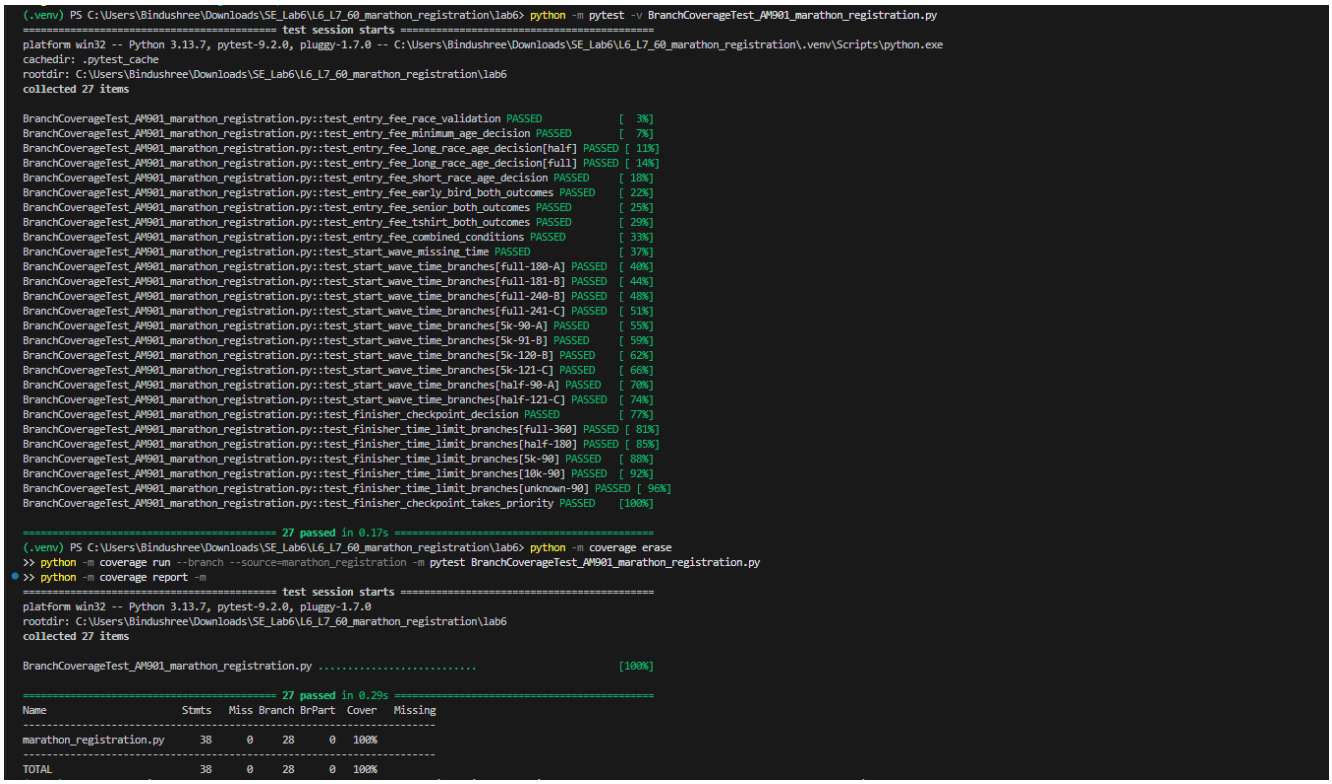
Command Used

python -m coverage erase
python -m coverage run --branch --source=marathon_registration -m pytest BranchCoverageTest_AM901_marathon_registration.py
python -m coverage report -m

Test file	Tests passed	Statements missed	Branches missed	Coverage
BranchCoverageTest_AM901_marathon_registration.py	27 / 27	0	0	100%

Result: All 27 branch-coverage tests passed. The report showed 38 statements, 0 missed statements, 28 branches, 0 partial branches, and 100% coverage.

After Result



7. Results Summary

Test stage	Tests passed	Coverage	Target met?
Original tests	2 / 2	Not measured in initial run	Tests passed
Code coverage tests	26 / 26	100% statement coverage	Yes
Branch coverage tests	27 / 27	100% coverage; 28 branches, 0 partial	Yes

8. Version Control and Commits

The provided terminal log shows four commits associated with this work. The two report/screenshot commits and the two test-file commits were pushed to origin/main.

- 5e90717 — Add code coverage test report
- f0a6262 — Add branch coverage test report
- 406090c — Add branch coverage tests
- 87fd1d4 — Add code coverage tests

The terminal output confirmed that git push origin main completed successfully. The .coverage data file remained untracked; it is generated by coverage.py and was not committed.

9. Conclusion

The original tests passed (2/2). The expanded code-coverage suite passed all 26 tests and achieved 100% statement coverage. The branch-coverage suite passed all 27 tests and achieved 100% coverage, with 28 branches and no partial branches reported. Both coverage results exceed the lab's 90% target.